92/100

CHE 454 - HW 2

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1. a) polystyrene in toluene

$MV = \text{molar volume} = \frac{\text{molar mass}}{\text{density}}$

$\text{number of segments} = \frac{MV \text{ polystyrene in toluene}}{MV \text{ toluene}}$ $\quad \frac{cm^3}{mol} \quad \frac{\frac{g}{mol}}{\frac{g}{cm^3}}$

$$= \frac{\frac{350{,}000 \frac{g}{mol}}{1.083 \frac{g}{cm^3}}}{\frac{92 \frac{g}{mol}}{0.861 \frac{g}{cm^3}}} = 3024.5 \approx \boxed{3025 \text{ segments}}$$ ✓

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b) polystyrene in MEK

same calculation $3560.1 \approx \boxed{3560 \text{ segments}}$ ✓

2. $\Delta S_m = -nR\left[X_A \ln X_A + X_B \ln X_B\right]$

a) $n = n_T + n_m$

$\frac{100g}{92 g/mol} = n_T = 1.087 mol$

$X_T = \frac{n_T}{T} = 0.439$ $\quad X_m = 1 - X_T$

$X_m = 0.561$

$R = 8.314 \frac{J}{mol\,K}$

$\frac{100g}{72 g/mol} = n_m = 1.389 mol$

Plug in

$n = 2.476 mol$

$$\boxed{\Delta S_m = 14.11 \frac{J}{K}}$$ ✓

b) $n = n_T + n_S$ $\quad n_T$ same as part a $\quad X_T = \frac{n_T}{n} = 0.999$

$n_S = \frac{100g}{1.25 \times 10^5 g/mol} = 8.33 \times 10^{-4} mol$

$X_S = 1 - X_T = 0.001$

$n = 1.088 mol$

b) cont

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$$\Phi_1 = \frac{n_1}{n_1 + X n_2} \qquad \Phi_2 = 1 - \Phi_1$$

$$\Delta S_m = -R(n_1 \ln \Phi_1 + n_2 \ln \Phi_2)$$

1 solvent
2 solute

$$X = \frac{\frac{92 g/mol}{0.861 g/cm^3}}{\frac{120,000 g/mol}{1.05 g/cm^3}} = 0.000935$$

why didn't you complete the answer? where is the value of deltaS?

c) $$\Phi_1 = \frac{x n_1}{x n_1 + x n_2} \qquad \Phi_2 = 1 - \Phi_1$$

$$\Delta S_m = -R(n_1 \ln \Phi_1 + n_2 \ln \Phi_2)$$

why didn't you complete the answer? where is the value of deltaS?

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Mixing toluene & MEK should give us a larger entropy shift - it should be the most prone to mixing

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3. a) $$\chi = \frac{1}{2} - \frac{A_2 M_1 d_2^2}{d_1}$$

solvent; 1.05 (polymer)

$$= \frac{1}{2} - \frac{0.000288 \frac{mol\,cm^3}{g^2} \cdot 98.96 \frac{g}{mol} \cdot 1.05 \frac{g}{cm^3} \cdot 1.05 \frac{g}{cm^3}}{1.24 \frac{g}{cm^3}}$$

$$= \boxed{0.475}$$

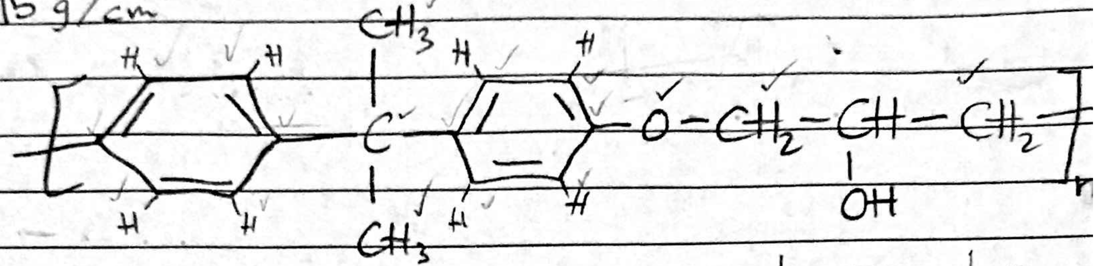
b) Same calculation w/ corresponding A, d1 and M1

$$\chi = \boxed{0.504}$$

$\boxed{\text{DCE is a better solvent, according to the calculations}}$

4. Calculate an estimate for solubility parameter

$\rho_p = 1.15\ g/cm^3$



$$\delta = \frac{\rho_p \sum f_i F_i}{M_0}$$

	F_i	f_i
(1) CH_2	131.5	2
(2) CH (aromatic)	117.12	8
(3) C (aromatic)	98.12	4
(4) 6 membered ring	-23.44	2
(5) CH_3	148.3	2
(6) C with no H	32.03	1
(7) >CH–	85.99	1
(8) –O– (ether)	114.98	1
(9) OH	225.84	1
(10) para substitution	40.33	2

$$M_0 = 18(12) + 2(16) + 20(1)$$
$$= 268\ g/mol$$

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$$\sum f_i F_i = 131.5(2) + 117.12(8) ...$$

$$\sum f_i F_i = 2381.66 \left(\frac{cal}{cc}\right)^{\frac{1}{2}} / mol$$

$$\delta = \frac{1.15 \cancel{\frac{g}{cm^3}} \left(2381.66 \left(\frac{cal}{cc}\right)^{\frac{1}{2}} / \cancel{mol}\right)}{268\ \cancel{g/mol}}$$

$$\boxed{\delta = 10.22 \left(\frac{cal}{cc}\right)^{\frac{1}{2}} / cm^3}$$

✓

5. Phase diagram for solvent-polymer system obeying the Flory-Huggins theory

$X_2 = 10{,}000$

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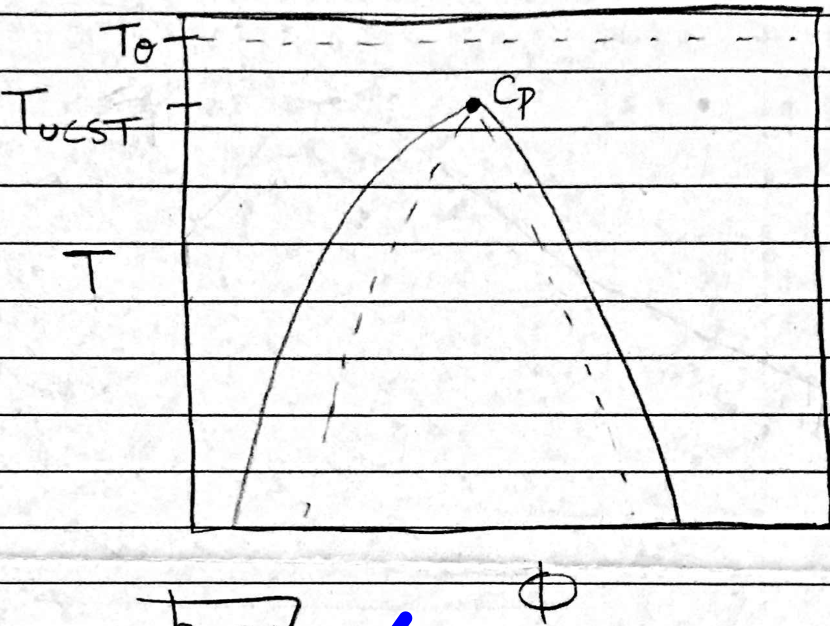

$$\Phi_{cr} = \frac{1}{\sqrt{10{,}000}} = \boxed{0.01}$$

$$\chi_{cr} = \frac{1}{2} + \frac{1}{\sqrt{10{,}000}} + \frac{1}{2(10{,}000)} = \boxed{0.51005}$$